

# Scientific Notes: Spatial Structure Analysis of Energy Patterns

**Objective:** Investigate the spatial structure and dynamical characteristics of Energy Pattern 1 (EP1) and Energy Pattern 2 (EP2) cyclones during intensification phase.

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## 1. Introduction

### 1.1 Motivation

Following the identification of three distinct energy patterns in South Atlantic cyclones through PCA-based K-Means clustering (see `cluster_analysis_energy_patterns/SCIENTIFIC_NOTES.md`), this analysis investigates the **spatial structure** of the most energetically active patterns.

**Energy Pattern Characteristics** (from cluster analysis): - **EP1 (11.6%)**: Large barotropic and baroclinic conversions; **exports energy** to surroundings - **EP2 (25.6%)**: Moderate balanced conversions; **imports energy** from large-scale environment  
- **EP3 (62.7%)**: Weak energetics representing typical “day-to-day” cyclones

#### **Rationale for EP1 and EP2 focus:**

EP3 cyclones exhibit weak energy budget activity and represent the climatological background. EP1 and EP2 cyclones, being energetically distinct and more intense, are the primary focus for understanding what structural characteristics differentiate high-impact systems from typical cyclones.

### 1.2 Research Questions

1. What are the spatial structures of baroclinic instability (EGR) in EP1 vs EP2?
2. How do upper-level (200 hPa) and low-level (850 hPa) PV anomalies differ between patterns?
3. What thermal advection patterns characterize each energy pattern?
4. How does near-surface moisture distribution and convergence differ?

**Approach:** Composite analysis of ERA5 reanalysis fields during the intensification phase of all EP1 (N=444) and EP2 (N=979) cyclones.

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## 2. Dataset

### 2.1 Sample Composition

| Energy Pattern | Cluster ID | Cases Analyzed | Percentage |
|----------------|------------|----------------|------------|
| <b>EP1</b>     | 0          | 444            | 11.6%      |
| <b>EP2</b>     | 2          | 979            | 25.6%      |

**Source:** All EP1 and EP2 cyclones from cluster analysis with complete lifecycle phases.

**Temporal Coverage:** Intensification phase only (6-hourly timesteps during deepening period)

## 2.2 ERA5 Reanalysis Data

**Variables Downloaded:** - **Pressure levels (hPa):** 175, 200, 225, 250, 500, 825, 850, 875, 975  
- **Pressure-level variables:** u, v, t, z, q (winds, temperature, geopotential, specific humidity) -  
**Single-level variables:** msl (mean sea level pressure)

**Spatial Configuration:** - **Resolution:**  $0.25^\circ \times 0.25^\circ$  - **Domain:**  $30^\circ \times 30^\circ$  centered on cyclone track center - **Composite domain:**  $15^\circ \times 15^\circ$  (marked in figures)

**Temporal Resolution:** 6-hourly

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## 3. Methodology

### 3.1 Diagnostic Fields Computed

| Diagnostic                            | Level(s)    | Purpose                                     | Key Reference                                  |
|---------------------------------------|-------------|---------------------------------------------|------------------------------------------------|
| <b>Eady Growth Rate</b>               | 500–850 hPa | Baroclinic instability measure              | Lindzen & Farrell (1980); Besson et al. (2021) |
| <b>Potential Vorticity</b>            | 200 hPa     | Upper-level dynamics, tropopause folding    | Hoskins et al. (1985)                          |
| <b>Potential Vorticity</b>            | 850 hPa     | Low-level PV anomaly, diabatic effects      | Čampa & Wernli (2012)                          |
| <b>Temperature Advection</b>          | 850 hPa     | Warm/cold advection patterns                | Sanders & Gyakum (1980)                        |
| <b>Specific Humidity</b>              | 975 hPa     | Near-surface moisture distribution          | -                                              |
| <b>Moisture Flux Divergence</b>       | 975 hPa     | Moisture convergence (convective potential) | Banacos & Schultz (2005)                       |
| <b>Sea Level Pressure</b>             | Surface     | Cyclone intensity and position              | -                                              |
| <b>RK criterion</b><br>(Rayleigh-Kuo) | 250 hPa     | Barotropic/baroclinic instability condition | Rayleigh (1880); Kuo (1949)                    |
| <b>KE Advection</b>                   | 250 hPa     | Kinetic energy tendency from advection      | -                                              |

|                                            |         |                                                       |                                                      |
|--------------------------------------------|---------|-------------------------------------------------------|------------------------------------------------------|
| <b>AFC</b> (Ageostrophic Flux Convergence) | 250 hPa | Eddy KE redistribution via ageostrophic pressure work | Orlanski & Katzfey (1991); Orlanski & Sheldon (1993) |
|--------------------------------------------|---------|-------------------------------------------------------|------------------------------------------------------|

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**Anomaly diagnostics** (departure from 1991–2020 climatology — same temporal decomposition as AFC):

| Anomaly Diagnostic               | Level   | Base Field Primed            | Output Variable |
|----------------------------------|---------|------------------------------|-----------------|
| <b>PV anomaly</b>                | 200 hPa | u , v , T at 175/200/225 hPa | pv_200_anom     |
| <b>PV anomaly</b>                | 850 hPa | u , v , T at 825/850/875 hPa | pv_850_anom     |
| <b>Temp advection anomaly</b>    | 850 hPa | u , v , T at 850 hPa         | adv_T_850_anom  |
| <b>Moisture flux div anomaly</b> | 975 hPa | u , v , q at 975 hPa         | div_q_975_anom  |
| <b>KE advection anomaly</b>      | 250 hPa | u , v at 250 hPa             | ke_adv_250_anom |
| <b>SLP anomaly</b>               | Surface | msl (Pa)                     | msl_anom        |

**Note:** EGR is not decomposed into anomaly form. The EGR formula involves  $N^2$  and vertical shear over a deep layer (500–850 hPa), making a clean temporal decomposition ill-defined. EGR is interpreted as a total-field baroclinic instability measure.

### 3.2 Spherical Grid Spacing

All horizontal derivatives account for Earth’s spherical geometry:

**Meridional spacing:**

$$dy = R_{\oplus} \Delta\phi$$

**Zonal spacing:**

$$dx = R_{\oplus} \cos(\phi) \Delta\lambda$$

where  $R_{\oplus} = 6.371 \times 10^6$  m,  $\phi$  = latitude,  $\lambda$  = longitude.

**Implementation:** `compute_spherical_grid_spacing(lat_1d, lon_1d)` - Verifies coordinate monotonicity

- Correct gradient signs regardless of coordinate direction - Uses `numpy.gradient()` for non-uniform grids

### 3.3 Eady Growth Rate (EGR)

**Formula:**

$$\sigma_{EGR} = 0.31 \frac{|f|}{N} \left| \frac{\partial \vec{V}}{\partial z} \right|$$

where: -  $f = 2\Omega \sin(\phi)$  = Coriolis parameter -  $N^2 = \frac{g}{\theta_v} \frac{\partial \theta_v}{\partial z}$  = static stability (Brunt-Väisälä frequency) -  $\left| \frac{\partial \vec{V}}{\partial z} \right|$  = vertical wind shear magnitude

**Layer:** 500–850 hPa (following Besson et al. 2021, Eq. 5: this layer encompasses the main lower-to-mid tropospheric baroclinic zone while avoiding contamination from the jet-level wind maximum at 250 hPa)

**Quality Control:** -  $N^2 > 10^{-6} \text{ s}^{-2}$  (exclude statically unstable regions) -  $|\phi| > 5^\circ$  (avoid equatorial singularity) -  $\sigma_{EGR} < 5 \text{ day}^{-1}$  (cap unrealistic values)

### 3.4 Potential Vorticity

**Formula:**

$$PV = -g(\zeta_\theta + f) \frac{\partial \theta}{\partial p}$$

where  $\zeta_\theta$  = relative vorticity on isentropic surface,  $\theta$  = potential temperature.

**Levels:** - **200 hPa:** Uses 175, 200, 225 hPa (upper-level tropopause dynamics) - **850 hPa:** Uses 825, 850, 875 hPa (low-level diabatic PV anomaly)

**Units:**  $\text{K m}^2 \text{ kg}^{-1} \text{ s}^{-1}$  (SI), converted to PVU ( $1 \text{ PVU} = 10^{-6} \text{ K m}^2 \text{ kg}^{-1} \text{ s}^{-1}$ ) in figures

### 3.5 Temperature Advection

**Formula:**

$$\text{adv}T = -\vec{V} \cdot \nabla T = -\left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y}\right)$$

**Level:** 850 hPa

**Sign Convention:** Positive = warm air advection; Negative = cold air advection

**Units:**  $\text{K h}^{-1}$  (converted from  $\text{K s}^{-1}$ )

### 3.6 Moisture Fields at 975 hPa

**Specific Humidity (q):** - Direct measure of atmospheric moisture content - Units:  $\text{g kg}^{-1}$  (converted from  $\text{kg kg}^{-1}$ ) - Level: 975 hPa (near-surface, above boundary layer)

**Moisture Flux Divergence:**

$$\nabla \cdot (q\vec{V}) = \frac{\partial(qu)}{\partial x} + \frac{\partial(qv)}{\partial y}$$

- **Negative values** → Moisture convergence (convective potential, latent heat release)
- **Positive values** → Moisture divergence (evaporation/drying)
- Units:  $\text{g kg}^{-1} \text{ h}^{-1}$  (converted from  $\text{kg kg}^{-1} \text{ s}^{-1}$ )
- Key diabatic process in cyclone intensification (Banacos & Schultz, 2005)

### 3.7 Rayleigh-Kuo Stability Criterion (250 hPa)

**Formula:**

$$RK = \beta - \frac{\partial^2 u}{\partial y^2}$$

where: -  $\beta = \frac{\partial f}{\partial y} = \frac{2\Omega \cos(\phi)}{R_\oplus} = \text{Rossby parameter}$  -  $\frac{\partial^2 u}{\partial y^2} = \text{meridional curvature of zonal wind}$



**Physical Interpretation:** - **Negative values** → Necessary condition for barotropic/baroclinic instability satisfied - **Combined criterion:** Incorporates planetary vorticity gradient and wind shear curvature - Level: 250 hPa (jet stream level, maximum barotropic effects)

**References:** Rayleigh (1880), Kuo (1949), Charney & Stern (1962)

### 3.8 Kinetic Energy Advection (250 hPa)

**Formula:**

$$KE_{adv} = -\vec{V} \cdot \nabla(KE) = -\vec{V} \cdot \nabla \left( \frac{1}{2}(u^2 + v^2) \right)$$

**Physical Interpretation:** - **Positive values** → KE advection causes local acceleration (energy gain) - **Negative values** → KE advection causes local deceleration (energy loss) - Level: 250 hPa (jet stream level) - Units:  $m^2 s^{-3}$

**Significance:**

Quantifies energy transport within the jet stream. Positive advection indicates regions where the flow pattern favors kinetic energy accumulation, potentially intensifying upper-level divergence and cyclone development.

### 3.9 Ageostrophic Flux Convergence — AFC (250 hPa)

The AFC diagnostic quantifies the redistribution of **eddy kinetic energy** through pressure work by the ageostrophic component of the eddy wind, following Orlanski & Katzfey (1991) and Orlanski & Sheldon (1993).

**Temporal decomposition (Solman & Menéndez 1998; Decker & Martin 2005):**

$$\vec{V} = \vec{V}_m + \vec{v}' \quad \Phi = \Phi_m + \phi'$$

where  $\vec{V}_m$ ,  $\Phi_m$  are the **30-year monthly climatological means** (1991–2020, WMO standard period) and  $\vec{v}'$ ,  $\phi'$  are the instantaneous eddy perturbations. This base state is independent of the area-mean decomposition used in the Lorenz Energy Cycle, providing an independent validation of the energy pathways.

**Geostrophic eddy wind:**

$$u'_g = -\frac{1}{f} \frac{\partial \phi'}{\partial y}, \quad v'_g = \frac{1}{f} \frac{\partial \phi'}{\partial x}$$

**Ageostrophic eddy wind:**

$$\vec{v}'_{ag} = \vec{v}' - \vec{v}'_g$$

**AFC (Ageostrophic Geopotential Flux Convergence):**

$$AFC = -\nabla \cdot (\vec{v}'_{ag} \phi')$$

**Sign convention:** - **Positive (AFC > 0):** Convergence of ageostrophic geopotential flux → **source** of eddy KE - **Negative (AFC < 0):** Divergence of flux → **sink** of eddy KE

**Units:**  $m^2 s^{-3}$  (  $W kg^{-1}$  )

**Climatology source:** ERA5 monthly averaged reanalysis (`reanalysis-era5-pressure-levels-monthly-means`), downloaded via CDS API. See `step2_1_download_era5_monthly_means.py`.

**References:** Orlanski & Katzfey (1991), Orlanski & Sheldon (1993), Solman & Menéndez (1998)

### 3.10 Anomaly Fields — Temporal Decomposition

To isolate the **synoptic-scale eddy signature** of EP1 and EP2 cyclones from the background climatological state, five additional diagnostics are computed as anomaly (eddy perturbation) fields using the same temporal decomposition already applied to AFC.

**Decomposition:**

$$X = \bar{X} + X'$$

where  $\bar{X}$  is the **30-year WMO monthly climatological mean** (1991–2020, ERA5 monthly-averaged reanalysis) and  $X'$  is the **eddy perturbation** at the time of the cyclone track.

**Climatology data source:**

ERA5 monthly-averaged reanalysis on pressure levels (`reanalysis-era5-pressure-levels-monthly-means`), downloaded via CDS API in `step2_1_download_era5_monthly_means.py`. The download is organized in four groups:

| Group  | Levels (hPa)  | Variables | Climatology file                        |
|--------|---------------|-----------|-----------------------------------------|
| 250hPa | 250           | u, v, z   | <code>era5_climatology_250hPa.nc</code> |
| pv200  | 175, 200, 225 | u, v, t   | <code>era5_climatology_pv200.nc</code>  |
| pv850  | 825, 850, 875 | u, v, t   | <code>era5_climatology_pv850.nc</code>  |
| mfd975 | 975           | u, v, q   | <code>era5_climatology_mfd975.nc</code> |
| slp    | surface       | msl       | <code>era5_climatology_slp.nc</code>    |

**Application to diagnostics:**

For each diagnostic  $D = D(u, v, T, \dots)$  the anomaly is computed by substituting all input fields with their eddy counterparts:

$$D' = D(u', v', T', \dots) \quad \text{where } u' = u - \bar{u}_m, \quad T' = T - \bar{T}_m, \dots$$

This is the standard linear perturbation approach in synoptic-scale diagnostic studies (e.g., Decker & Martin 2005). For non-linear diagnostics (PV, divergence), the anomaly field thus captures the contribution of synoptic-scale transients to the diagnostic.

**Physical rationale:**

- **PV** : Highlights stratospheric intrusions and diabatic generation that are anomalous relative to the climatological background tropopause structure.
- **Temp advection** : Isolates warm/cold advection driven by the cyclone’s eddy circulation, removing the climatological advective background.
- **Moisture flux div** : Captures synoptic-scale convergence anomalies, filtering out the seasonal moisture cycle.
- **KE advection** : Highlights energy transport by the eddy jet/streak relative to the mean flow.

**Note on EGR:** EGR is a layer-averaged diagnostic derived from the total wind shear and static stability. A temporal decomposition of EGR would require priming  $N^2$  and the shear simultaneously, leading to non-trivial cross terms. For this reason EGR is retained as a total-field diagnostic only.

**Note on AFC:** AFC is by construction an anomaly field (it is already computed from eddy  $\vec{v}'$  and  $\phi'$  relative to the monthly climatology). No additional anomaly version is needed.

## 4. Results

**Note:** Statistical summaries presented below are computed from spatial composites centered on cyclone genesis locations. Each subsection presents the composite figure followed by quantitative statistics. Physical interpretation of spatial patterns and EP1 vs EP2 differences is currently under analysis.

### 4.1 Eady Growth Rate (Baroclinic Instability)

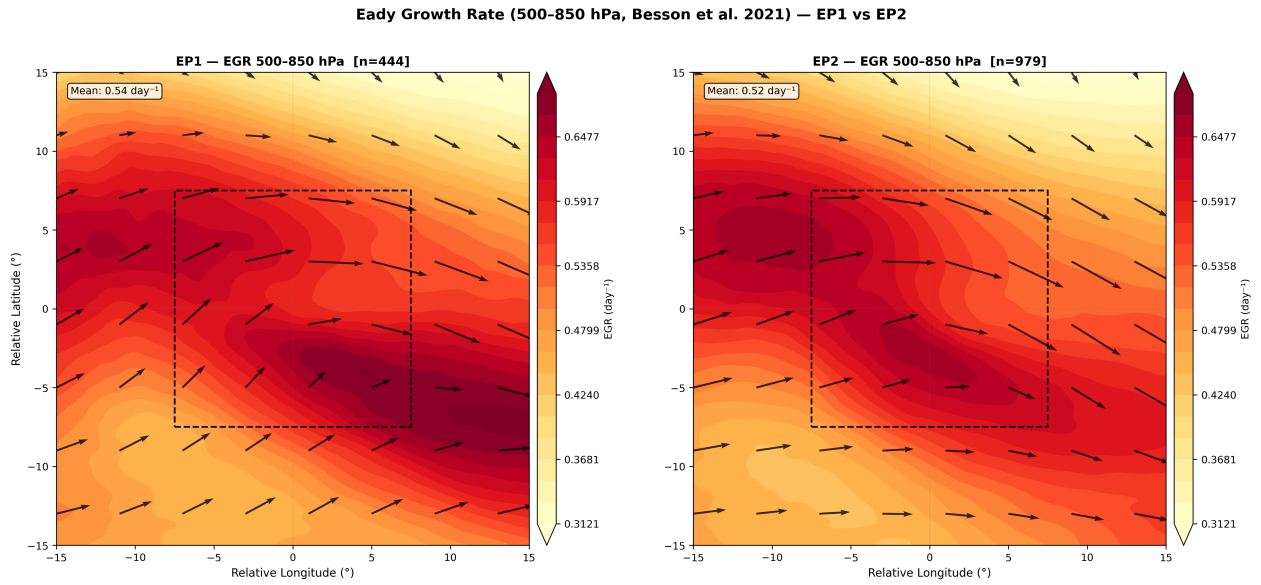


Figure 1: EGR Composite

Figure: Eady growth rate composite (500–850 hPa layer, Besson et al. 2021) for EP1 (left) and EP2 (right). Contours show growth rate in  $\text{day}^{-1}$ . The  $15^\circ \times 15^\circ$  box marks the LEC computation domain.

#### Summary Statistics:

| Statistic                           | EP1             | EP2             |
|-------------------------------------|-----------------|-----------------|
| Mean $\pm$ SD ( $\text{day}^{-1}$ ) | $0.56 \pm 0.09$ | $0.56 \pm 0.10$ |
| Median ( $\text{day}^{-1}$ )        | 0.57            | 0.58            |
| Range ( $\text{day}^{-1}$ )         | [0.37, 0.73]    | [0.36, 0.72]    |

### 4.2 Upper-Level Dynamics (200 hPa PV)

Figure: Potential vorticity at 200 hPa for EP1 (left) and EP2 (right). Units: PVU ( $10^{-6} \text{ K m}^2 \text{ kg}^{-1} \text{ s}^{-1}$ ). Shows tropopause dynamics and stratospheric intrusions.

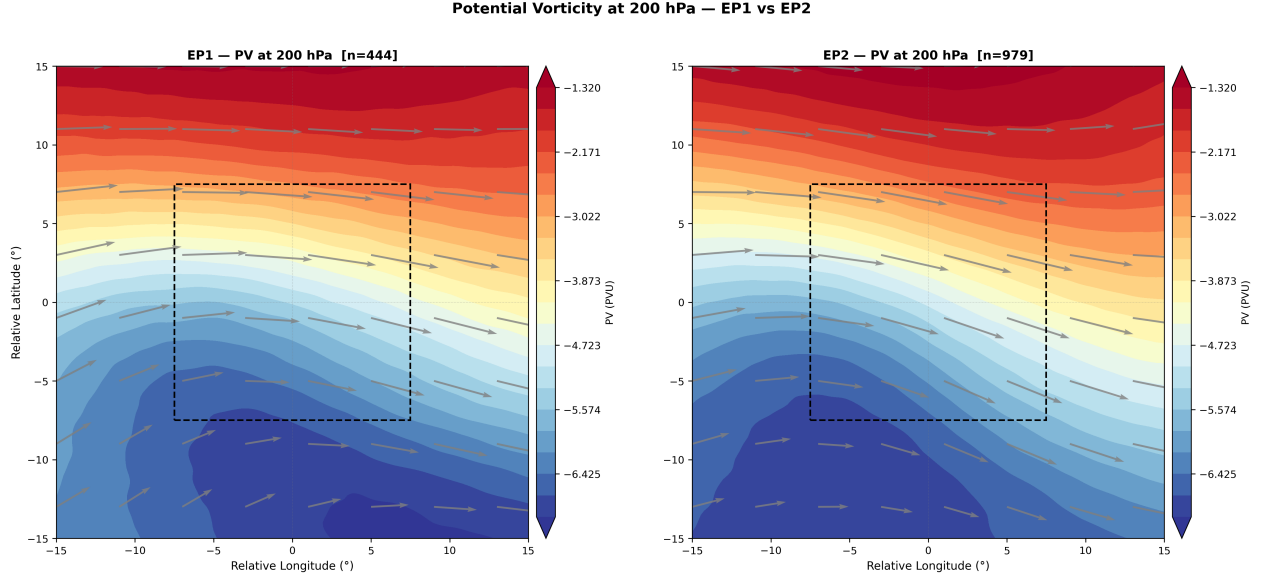


Figure 2: PV@200 Composite

#### Summary Statistics:

| Statistic   | EP1            | EP2            |
|-------------|----------------|----------------|
| Mean (PVU)  | -4.54          | -4.42          |
| Range (PVU) | [-7.08, -1.26] | [-6.86, -1.18] |

#### 4.3 Low-Level PV Anomaly (850 hPa)

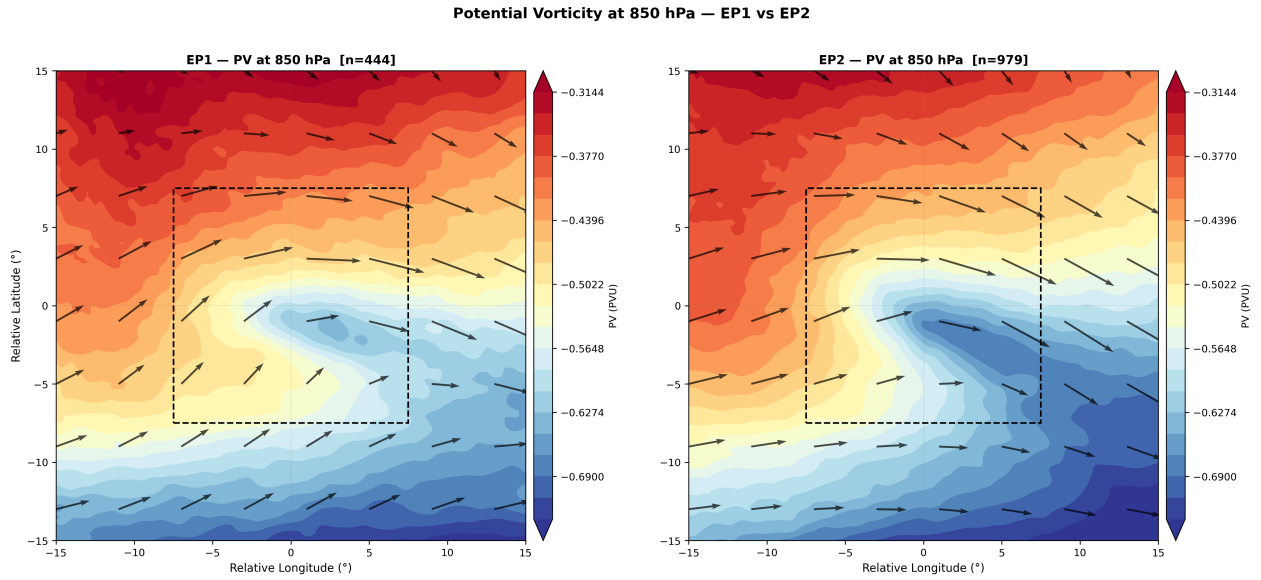


Figure 3: PV@850 Composite

*Figure: Potential vorticity at 850 hPa for EP1 (left) and EP2 (right). Units: PVU. Indicates diabatic PV generation and low-level cyclonic circulation.*

#### Summary Statistics:

| Statistic | EP1 | EP2 |
|-----------|-----|-----|
|-----------|-----|-----|

| Statistic                                            | EP1    | EP2    |
|------------------------------------------------------|--------|--------|
| Max cold advection ( $\text{K h}^{-1}$ )             | -0.132 | -0.119 |
| LEC $15 \times 15^\circ$ mean ( $\text{K h}^{-1}$ )  | -0.070 | -0.021 |
| Full $30 \times 30^\circ$ mean ( $\text{K h}^{-1}$ ) | -0.031 | -0.003 |

#### Lateral boundaries (flux assessment):

| Boundary               | EP1    | EP2    |
|------------------------|--------|--------|
| North ( $+7.5^\circ$ ) | -0.025 | 0.020  |
| South ( $-7.5^\circ$ ) | -0.081 | -0.037 |
| East ( $+7.5^\circ$ )  | -0.008 | 0.068  |
| West ( $-7.5^\circ$ )  | -0.095 | -0.074 |

#### 4.5 Sea Level Pressure

Figure: Mean sea level pressure for EP1 (left) and EP2 (right). Units: hPa. Shows cyclone intensity and horizontal structure.

##### Summary Statistics:

| Statistic                            | EP1    | EP2    |
|--------------------------------------|--------|--------|
| Minimum (hPa)                        | 996.5  | 994.8  |
| Maximum (hPa)                        | 1018.3 | 1019.2 |
| LEC $15 \times 15^\circ$ mean (hPa)  | 1008.1 | 1005.2 |
| Full $30 \times 30^\circ$ mean (hPa) | 1009.3 | 1006.4 |

#### 4.6 Specific Humidity (975 hPa)

Figure: Specific humidity (shading,  $\text{g kg}^{-1}$ ) and moisture flux vectors at 975 hPa for EP1 (left) and EP2 (right).

##### Summary Statistics:

| Statistic                                             | EP1           | EP2           |
|-------------------------------------------------------|---------------|---------------|
| Mean ( $\text{g kg}^{-1}$ )                           | 6.37          | 6.52          |
| Range ( $\text{g kg}^{-1}$ )                          | [2.79, 11.14] | [3.13, 11.16] |
| LEC $15 \times 15^\circ$ mean ( $\text{g kg}^{-1}$ )  | 6.31          | 6.58          |
| Full $30 \times 30^\circ$ mean ( $\text{g kg}^{-1}$ ) | 6.37          | 6.52          |

#### 4.7 Rayleigh-Kuo Stability Criterion (250 hPa)

Figure: Rayleigh-Kuo criterion at 250 hPa for EP1 (left) and EP2 (right). Units:  $\text{s}^{-1}$ . Negative values (blue) indicate regions satisfying the necessary condition for barotropic/baroclinic instability. Wind vectors show 250 hPa flow.

##### Summary Statistics:

### Temperature Advection at 850 hPa — EP1 vs EP2

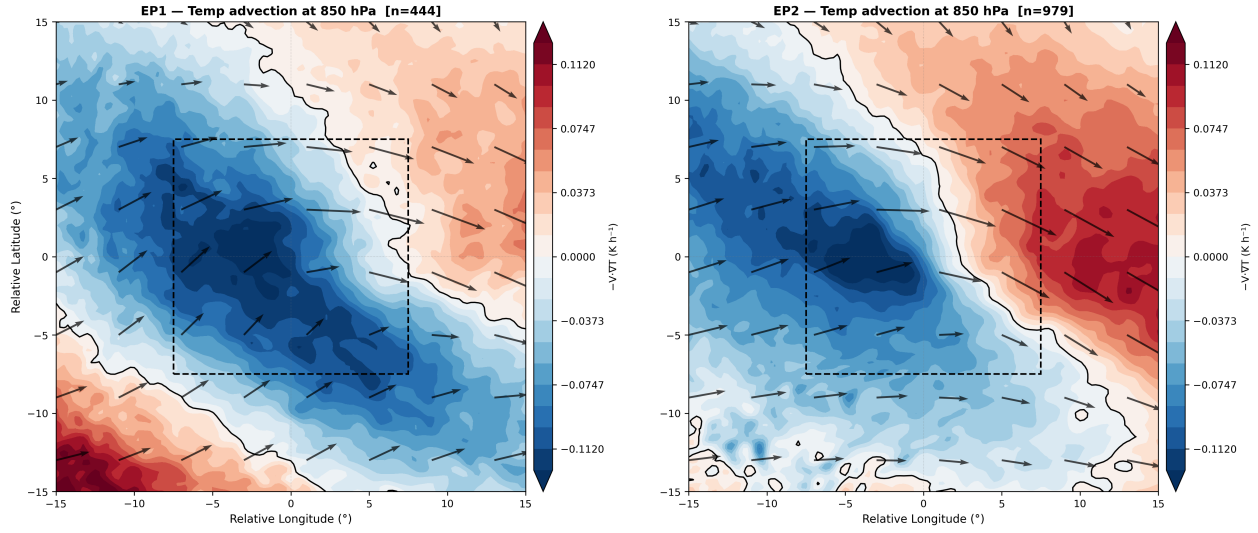


Figure 4: Temp Advection Composite

### Sea Level Pressure — EP1 vs EP2

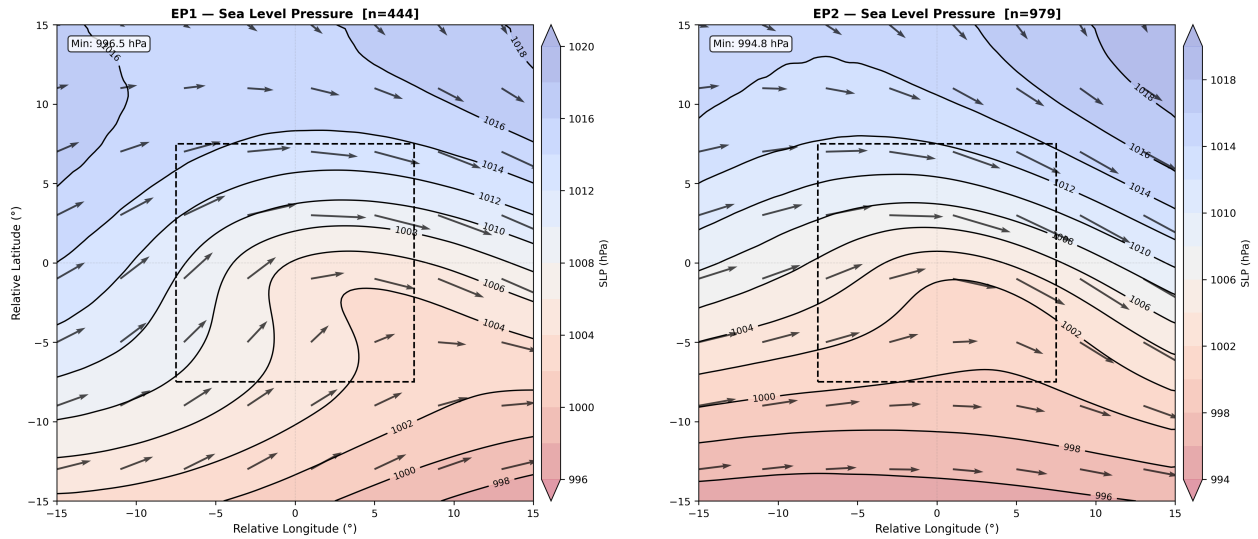


Figure 5: SLP Composite

# Low-Level Moisture Transport at 975 hPa — EP1 vs EP2

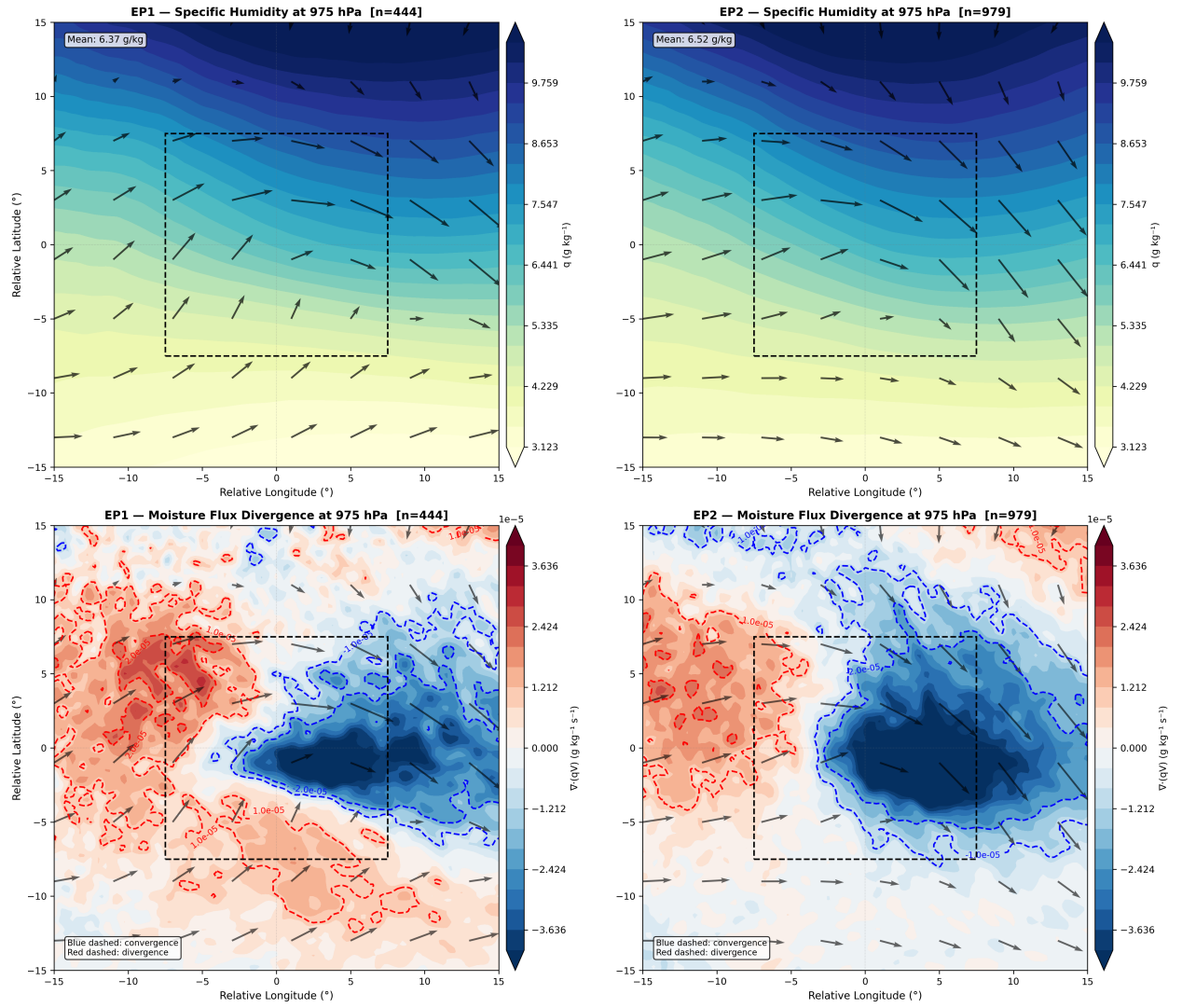


Figure 6: Moisture Composite

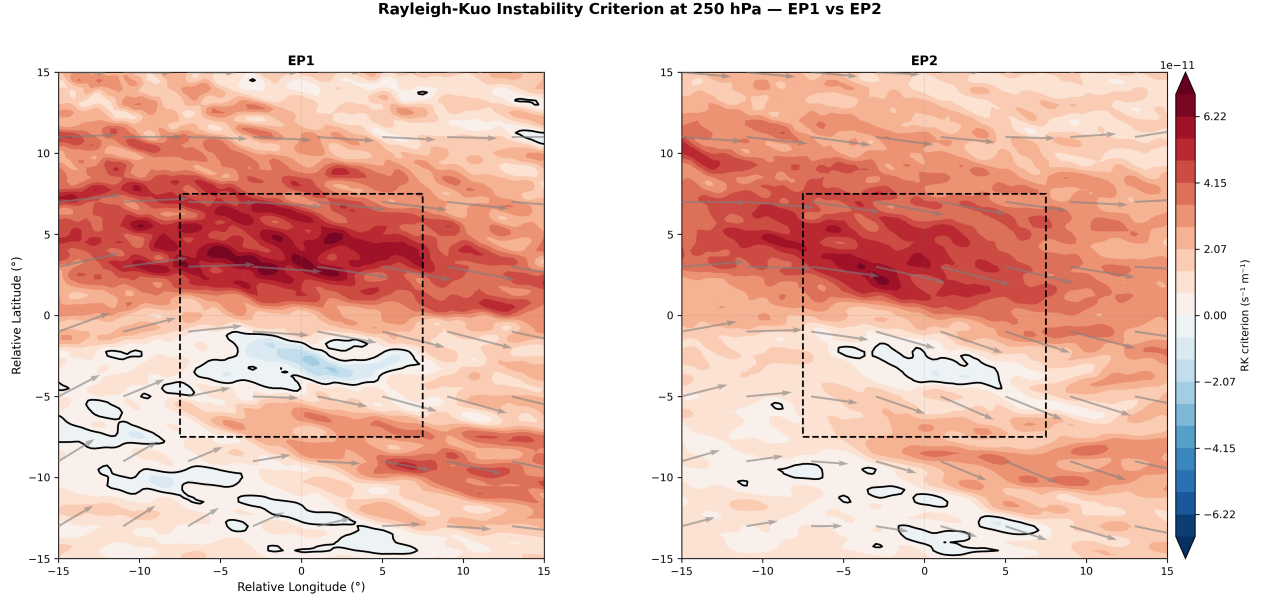


Figure 7: RK Criterion Composite

| Statistic                      | EP1                     | EP2                     |
|--------------------------------|-------------------------|-------------------------|
| Mean ( $s^{-1}$ )              | 2.242e-11               | 2.271e-11               |
| Range ( $s^{-1}$ )             | [-2.427e-11, 6.908e-11] | [-7.869e-12, 6.445e-11] |
| LEC $15 \times 15^\circ$ mean  | 2.814e-11               | 2.818e-11               |
| Full $30 \times 30^\circ$ mean | 2.242e-11               | 2.271e-11               |

*Negative values indicate regions satisfying the necessary condition for barotropic/baroclinic instability.*

#### 4.8 Kinetic Energy Advection (250 hPa)

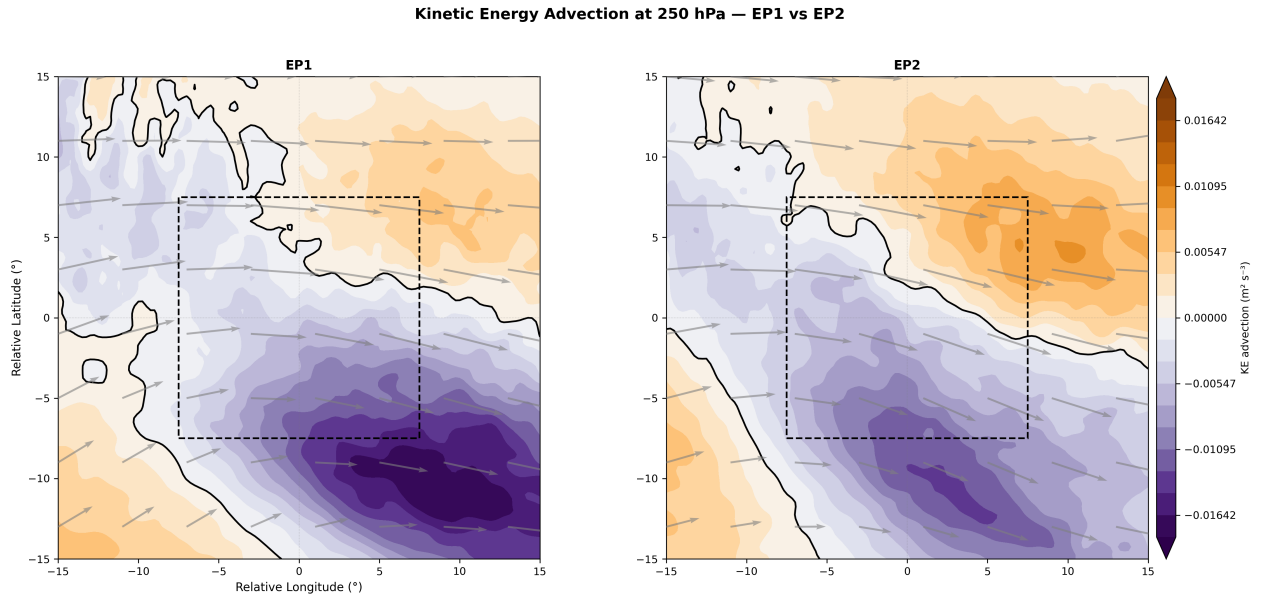


Figure 8: KE Advection Composite

*Figure: Kinetic energy advection at 250 hPa for EP1 (left) and EP2 (right). Units:  $m^2 s^{-3}$ . Positive values (purple) indicate KE gain through advection; negative values (orange) indicate KE loss. Wind*



| Boundary      | EP1        | EP2        |
|---------------|------------|------------|
| North (+7.5°) | 1.207e-03  | 3.303e-03  |
| South (-7.5°) | -7.365e-03 | -5.867e-03 |
| East (+7.5°)  | -1.957e-03 | 1.570e-03  |
| West (-7.5°)  | -1.308e-03 | -1.924e-03 |

#### 4.9 Ageostrophic Flux Convergence (250 hPa)

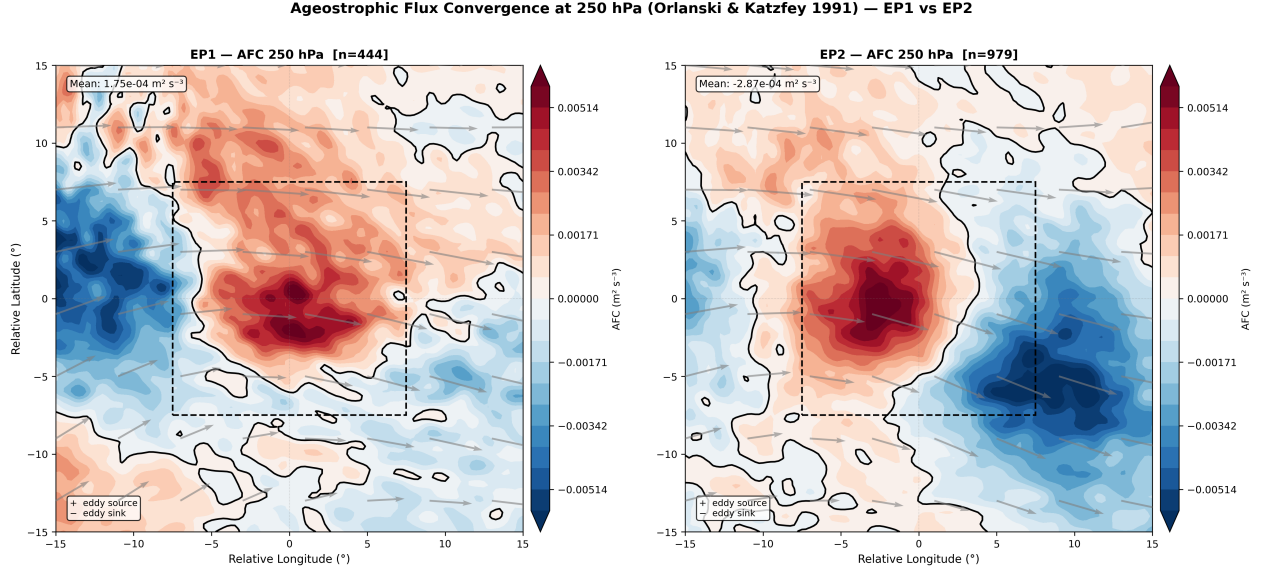


Figure 9: AFC Composite

Figure: Ageostrophic Flux Convergence at 250 hPa for EP1 (left) and EP2 (right). Units:  $m^2 s^{-3}$ . Positive values (red) indicate eddy KE sources; negative values (blue) indicate eddy KE sinks. Base state: ERA5 30-year monthly climatology (1991–2020). Wind vectors show eddy perturbation wind  $\vec{v}'$  at 250 hPa.

#### Summary Statistics:

| Statistic              | EP1                     | EP2                     |
|------------------------|-------------------------|-------------------------|
| Mean ( $m^2 s^{-3}$ )  | 1.752e-04               | -2.870e-04              |
| Range ( $m^2 s^{-3}$ ) | [-6.170e-03, 6.491e-03] | [-6.673e-03, 6.259e-03] |
| LEC 15×15° mean        | 1.575e-03               | 7.611e-04               |
| Full 30×30° mean       | 1.752e-04               | -2.870e-04              |

#### Lateral boundaries (flux assessment):

| Boundary      | EP1        | EP2        |
|---------------|------------|------------|
| North (+7.5°) | 2.092e-03  | 5.479e-04  |
| South (-7.5°) | -5.042e-04 | -1.389e-03 |

| Boundary     | EP1        | EP2        |
|--------------|------------|------------|
| East (+7.5°) | 6.667e-04  | -3.241e-03 |
| West (-7.5°) | -1.856e-03 | 1.532e-03  |

#### 4.10 PV Anomaly at 200 hPa

Figure: PV eddy perturbation at 200 hPa (departure from 1991–2020 climatology) for EP1 (left) and EP2 (right). Units: PVU. Computed from eddy winds  $u$ ,  $v$  and temperature  $T$  at 175/200/225 hPa.

##### Summary Statistics:

| Statistic                | EP1 | EP2 |
|--------------------------|-----|-----|
| Mean $\pm$ SD (PVU)      | TBD | TBD |
| Range (PVU)              | TBD | TBD |
| LEC 15 $\times$ 15° mean | TBD | TBD |

#### 4.11 PV Anomaly at 850 hPa

Figure: PV eddy perturbation at 850 hPa (departure from 1991–2020 climatology) for EP1 (left) and EP2 (right). Units: PVU. Computed from eddy winds  $u$ ,  $v$  and temperature  $T$  at 825/850/875 hPa.

##### Summary Statistics:

| Statistic                | EP1 | EP2 |
|--------------------------|-----|-----|
| Mean $\pm$ SD (PVU)      | TBD | TBD |
| Range (PVU)              | TBD | TBD |
| LEC 15 $\times$ 15° mean | TBD | TBD |

#### 4.12 Temperature Advection Anomaly (850 hPa)

Figure: Temperature advection eddy perturbation at 850 hPa ( $-V \cdot T$ ) for EP1 (left) and EP2 (right). Units:  $K h^{-1}$ . Positive = anomalous warm advection; Negative = anomalous cold advection.

##### Summary Statistics:

| Statistic                               | EP1 | EP2 |
|-----------------------------------------|-----|-----|
| Domain mean ( $K h^{-1}$ )              | TBD | TBD |
| Max warm advection ( $K h^{-1}$ )       | TBD | TBD |
| Max cold advection ( $K h^{-1}$ )       | TBD | TBD |
| LEC 15 $\times$ 15° mean ( $K h^{-1}$ ) | TBD | TBD |

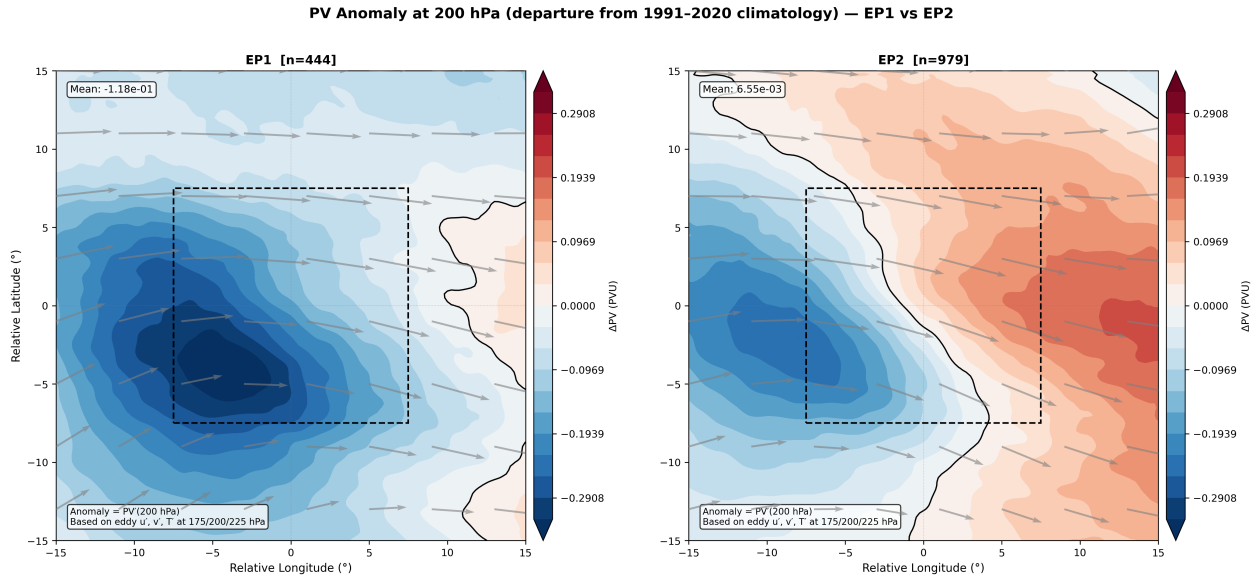


Figure 10: PV @200 Composite

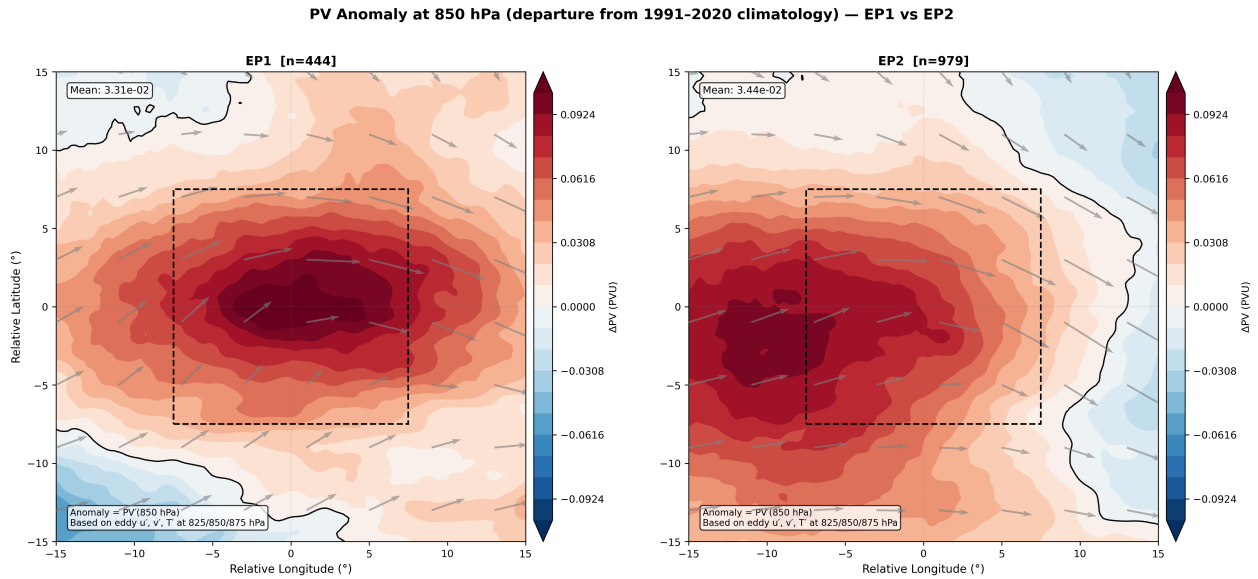


Figure 11: PV @850 Composite

Temperature Advection Anomaly at 850 hPa (departure from climatology) — EP1 vs EP2

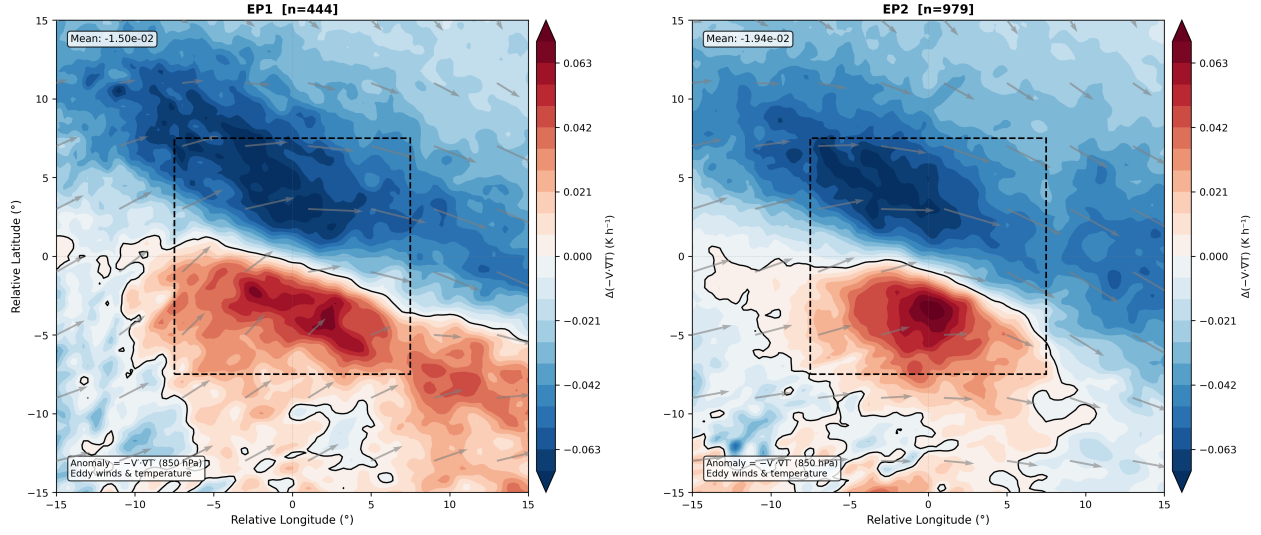


Figure 12: AdvT @850 Composite

Moisture Flux Divergence Anomaly at 975 hPa (departure from climatology) — EP1 vs EP2

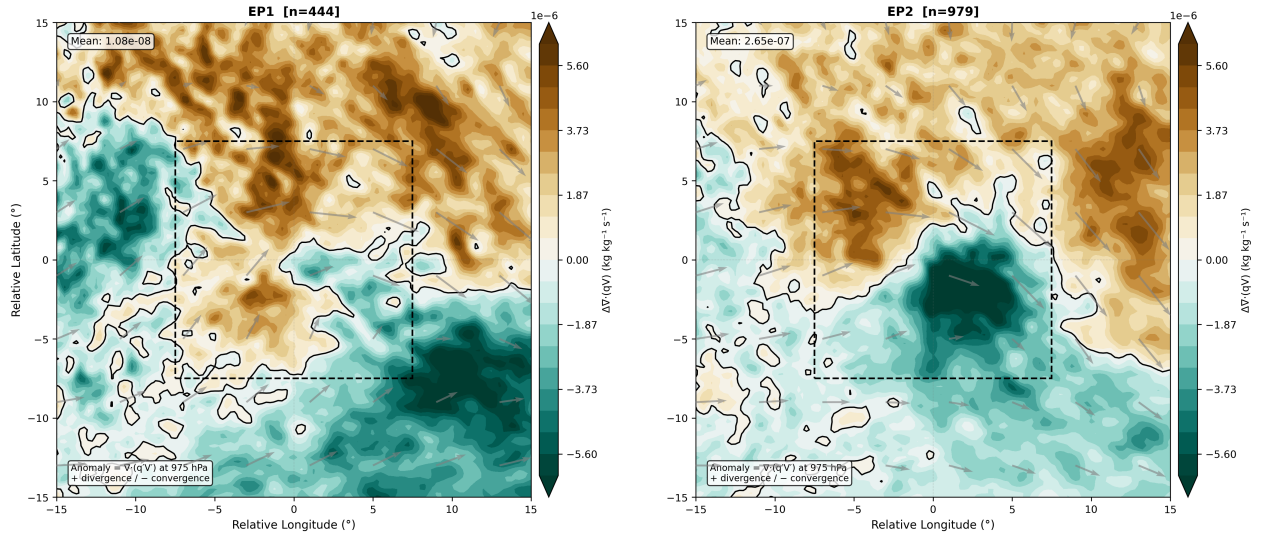


Figure 13: MFD @975 Composite

#### 4.13 Moisture Flux Divergence Anomaly (975 hPa)

Figure: Moisture flux divergence eddy perturbation at 975 hPa ( $\nabla \cdot (q V)$ ) for EP1 (left) and EP2 (right). Units:  $\text{kg kg}^{-1} \text{s}^{-1}$ . Negative = anomalous convergence.

Summary Statistics:

| Statistic                                         | EP1 | EP2 |
|---------------------------------------------------|-----|-----|
| Domain mean ( $\text{kg kg}^{-1} \text{s}^{-1}$ ) | TBD | TBD |
| Max convergence anomaly                           | TBD | TBD |
| LEC $15 \times 15^\circ$ mean                     | TBD | TBD |

#### 4.14 KE Advection Anomaly (250 hPa)

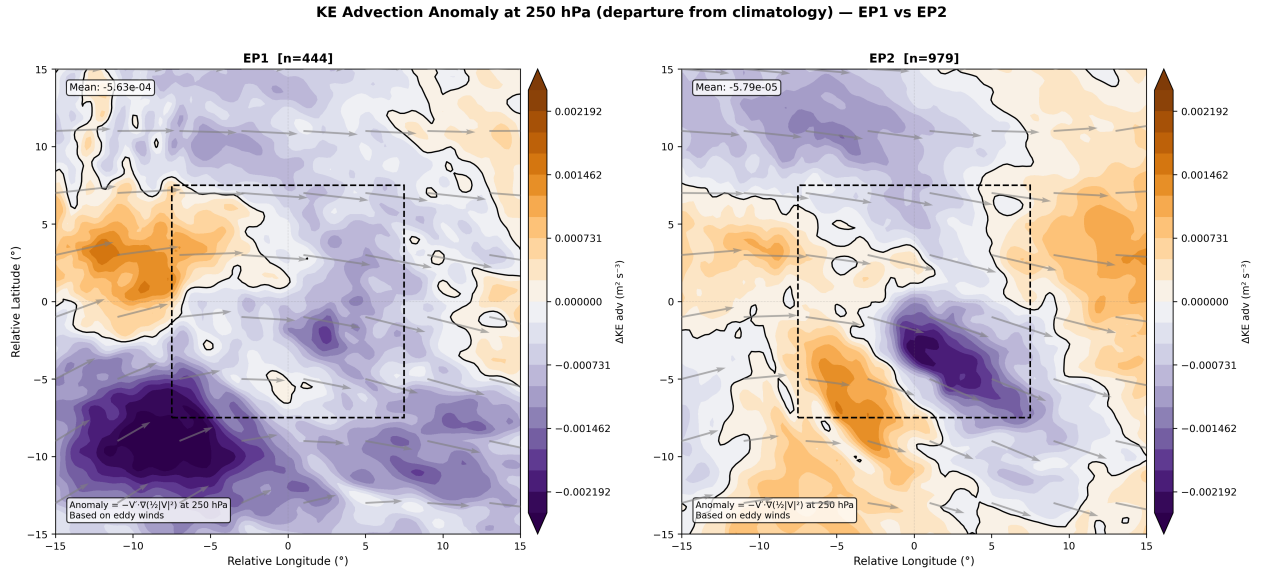


Figure 14: KE Adv@250 Composite

Figure: Kinetic energy advection eddy perturbation at 250 hPa ( $-V \cdot (\frac{1}{2}|V|^2)$ ) for EP1 (left) and EP2 (right). Units:  $\text{m}^2 \text{s}^{-3}$ .

Summary Statistics:

| Statistic                                  | EP1 | EP2 |
|--------------------------------------------|-----|-----|
| Domain mean ( $\text{m}^2 \text{s}^{-3}$ ) | TBD | TBD |
| Range ( $\text{m}^2 \text{s}^{-3}$ )       | TBD | TBD |
| LEC $15 \times 15^\circ$ mean              | TBD | TBD |

#### 4.15 Sea Level Pressure Anomaly

Figure: Sea level pressure anomaly ( $SLP = SLP - \text{climatological monthly mean}$ ) for EP1 (left) and EP2 (right). Units: hPa. Positive (red) = anomalous high pressure; Negative (blue) = anomalous low pressure. The low-pressure anomaly centred at the cyclone position directly indicates the deepening relative to the climatological background. 850 hPa wind vectors overlaid.

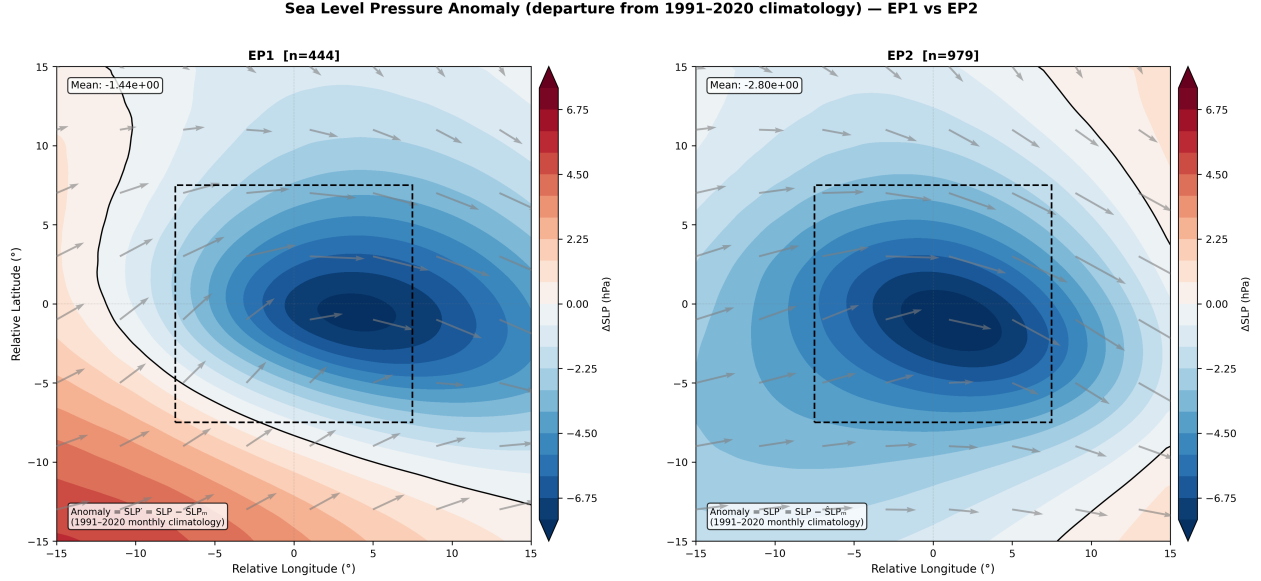


Figure 15: SLP Composite

### Summary Statistics:

| Statistic                           | EP1 | EP2 |
|-------------------------------------|-----|-----|
| Domain mean (hPa)                   | TBD | TBD |
| Minimum anomaly (hPa)               | TBD | TBD |
| LEC $15 \times 15^\circ$ mean (hPa) | TBD | TBD |

## 5. Physical Interpretation

**Status:** Analysis of spatial patterns and physical mechanisms is ongoing. Preliminary observations include:

### 5.1 EP1 Characteristics

- {EP1\_INTERPRETATION}

### 5.2 EP2 Characteristics

- {EP2\_INTERPRETATION}

### 5.3 Key Structural Differences

- {KEY\_DIFFERENCES}

**Future Work:** Detailed analysis of spatial patterns, vertical coupling mechanisms, and relationship between energetics (from cluster analysis) and structural characteristics.

## 6. Computational Implementation

### 6.1 Grid Spacing on Spherical Earth

**Function:** `compute_spherical_grid_spacing(lat_1d, lon_1d)`

Computes accurate grid spacing accounting for Earth's spherical geometry:

```
# Meridional spacing (constant per latitude band)
dy = R_EARTH * Δ # meters

# Zonal spacing (varies with latitude)
dx = R_EARTH * cos( ) * Δ # meters
```

**Quality checks:** - Verifies latitude is monotonic (increasing or decreasing) - Verifies longitude is monotonic - Raises `ValueError` if non-monotonic - Gradient sign automatically correct regardless of coordinate direction

### 6.2 Divergence on Spherical Coordinates

For small domains ( $\sim 30^\circ$ ), the simplified Cartesian approximation is valid:

$$\nabla \cdot \vec{F} \approx \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y}$$

where  $dx$  and  $dy$  account for spherical geometry (Section 6.1).

**Full spherical formula** (not required for  $30^\circ$  domain):

$$\nabla \cdot \vec{F} = \frac{1}{R \cos \phi} \frac{\partial F_\lambda}{\partial \lambda} + \frac{1}{R \cos \phi} \frac{\partial (F_\phi \cos \phi)}{\partial \phi}$$

Our implementation uses the simplified form with accurate  $dx$ ,  $dy$  — appropriate for mesoscale domains.

### 6.3 Unit Tracking

All calculations use **MetPy units** for automatic dimensional analysis:

```
# Example: moisture flux
qu = (q * units('kg/kg')) * (u * units('m/s'))
# → qu.units = 'kg / kg * m / s' (automatically tracked)

# Convert divergence to g/kg/s
div_q_si = (dqu_dx + dqv_dy) * units('kg/kg/s')
div_q_gkg = (div_q_si * 1000 * units('g/kg')).magnitude
```

**Benefits:** - Prevents unit errors (e.g., mixing K and  $^\circ\text{C}$ ) - Documents unit transformations  
- Explicit conversions (no hardcoded magic numbers)

## 6.4 Numerical Methods

- **Spatial derivatives:** `numpy.gradient()` with 2nd-order centred finite differences
- **Vertical derivatives:** Centred FD over 3 levels (e.g., 175–200–225 hPa for PV@200)
- **Interpolation:** Linear interpolation to regular  $0.25^\circ$  grid via `xarray.interp()`

## 6.5 Quality Control Filters

| Diagnostic | Filter                              | Rationale                               |
|------------|-------------------------------------|-----------------------------------------|
| EGR        | $N^2 > 10^{-6} \text{ s}^{-2}$      | Exclude statically unstable regions     |
| EGR        | $\ \phi\  > 5^\circ$                | Avoid near-equatorial ( $f \approx 0$ ) |
| EGR        | $\sigma_{EGR} < 5 \text{ day}^{-1}$ | Cap unrealistic growth rates            |
| PV         | Valid only where $\theta$ monotonic | Ensure isentropic sorting               |

## 6.6 Code Verification

**Test cases:** 1. Zonal wind  $\rightarrow$  zero meridional derivative 2. Meridional wind  $\rightarrow$  zero zonal derivative  
 3. Uniform field  $\rightarrow$  zero divergence 4. Reversed latitude ( $90 \rightarrow -90$ )  $\rightarrow$  correct gradient sign

**Validation:** - Cross-check EGR values with literature (typical:  $0.3\text{--}1.0 \text{ day}^{-1}$ ) - PV@200 should align with tropopause ( $\sim 2\text{--}3$  PVU contour) - Compare temperature advection with operational analyses

## 7. References

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